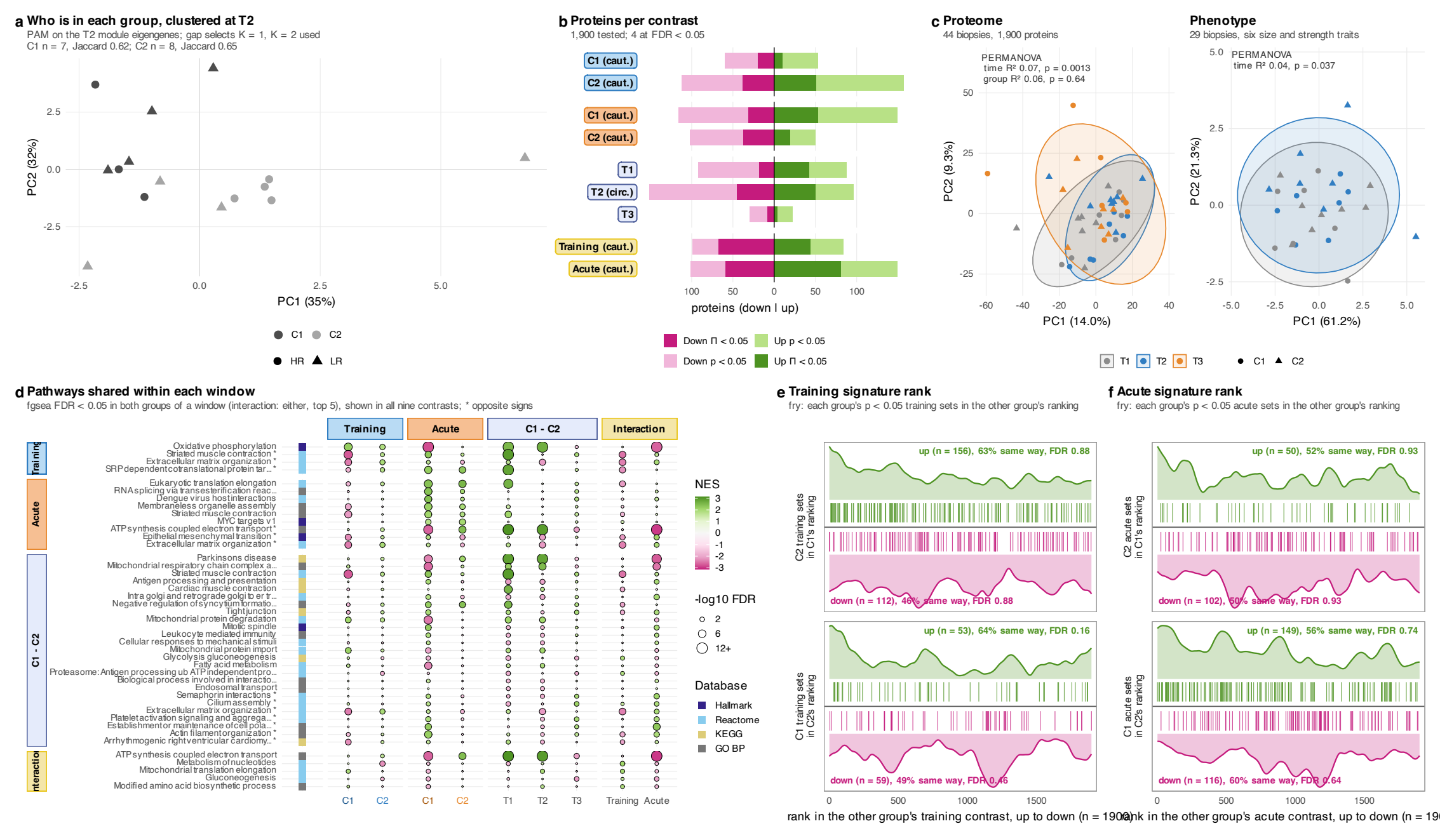

a Who is in each group, clustered at T2
PAM on the T2 module eigengenes; gap selects K = 1, K = 2 used
C1 n = 7, Jaccard 0.62; C2 n = 8, Jaccard 0.65
PC2 (32%)
PC1 (35%)
C1
C2
HR
LR
b Proteins per contrast
1,900 tested; 4 at FDR < 0.05
C1 (caut.)
C2 (caut.)
C1 (caut.)
C2 (caut.)
T1
T2 (circ.)
T3
Training (caut.)
Acute (caut.)
proteins (down | up)
Down Π < 0.05
Up p < 0.05
Down p < 0.05
Up Π < 0.05
c Proteome
44 biopsies, 1,900 proteins
PERMANOVA
time R² 0.07, p = 0.0013
group R² 0.06, p = 0.64
PC2 (9.3%)
PC1 (14.0%)
Phenotype
29 biopsies, six size and strength traits
PERMANOVA
time R² 0.04, p = 0.037
PC2 (21.3%)
PC1 (61.2%)
T1
T2
T3
C1
C2
d Pathways shared within each window
fgsea FDR < 0.05 in both groups of a window (interaction: either, top 5), shown in all nine contrasts; * opposite signs
Training
Acute
C1 - C2
Interaction
Oxidative phosphorylation
Striated muscle contraction *
Extracellular matrix organization *
SRP dependent cotranslational protein tar... *
Eukaryotic translation elongation
RNA splicing via transesterification reac...
Dengue virus host interactions
Membraneless organelle assembly
Striated muscle contraction
MYC targets v1
ATP synthesis coupled electron transport *
Epithelial mesenchymal transition *
Extracellular matrix organization *
Parkinsons disease
Mitochondrial respiratory chain complex a...
Striated muscle contraction
Antigen processing and presentation
Cardiac muscle contraction
Intra golgi and retrograde golgi to er tr...
Negative regulation of syncytium formatio...
Tight junction
Mitochondrial protein degradation
Mitotic spindle
Leukocyte mediated immunity
Cellular responses to mechanical stimuli
Mitochondrial protein import
Glycolysis gluconeogenesis
Fatty acid metabolism
Proteasome: Antigen processing ub ATP independent pro...
Biological process involved in interactio...
Endosomal transport
Semaphorin interactions *
Cilium assembly *
Extracellular matrix organization *
Platelet activation signaling and aggrega... *
Establishment or maintenance of cell pola... *
Actin filament organization *
Arrhythmogenic right ventricular cardiomy... *
ATP synthesis coupled electron transport
Metabolism of nucleotides
Mitochondrial translation elongation
Gluconeogenesis
Modified amino acid biosynthetic process
C1
C2
C1
C2
T1
T2
T3
Training
Acute
NES
3
2
1
0
-1
-2
-3
-log10 FDR
2
6
12+
Database
Hallmark
Reactome
KEGG
GO BP
e Training signature rank
fry: each group's p < 0.05 training sets in the other group's ranking
C2 training sets in C1's ranking
up (n = 156), 63% same way, FDR 0.88
down (n = 112), 46% same way, FDR 0.88
C1 training sets in C2's ranking
up (n = 53), 64% same way, FDR 0.16
down (n = 59), 49% same way, FDR 0.16
0
500
1000
1500
rank in the other group's training contrast, up to down (n = 1900)
f Acute signature rank
fry: each group's p < 0.05 acute sets in the other group's ranking
C2 acute sets in C1's ranking
up (n = 50), 52% same way, FDR 0.93
down (n = 102), 50% same way, FDR 0.93
C1 acute sets in C2's ranking
up (n = 149), 56% same way, FDR 0.74
down (n = 116), 60% same way, FDR 0.64
0
500
1000
1500
rank in the other group's acute contrast, up to down (n = 19